



UHMWPE 2000



Ultra-high molecular weight low pressure polyethylene with a molecular weight > 9.000.000 g/mol plus additives for the improvement of specific material properties.

Our UHMWPE 2000 grade high performance plastics are developed for very special and demanding applications which are beyond the boundaries of the normal 1000 grade materials.

Grades available

UHMWPE 2000 – AB 995

Anti-microbial is a UHMW Polymer. It has excellent anti-microbial protection. The protection range which has been measured is 99,99 % against bacteria "Staphylococcus aureus ATCC6538P" and "Klebsiella Pneumoniae ATCC 4352". This grade is FDA compliant for food industry.

UHMWPE 2000 - #588 Blue

By adding a special filler we can improve the wear resistance to achieve an even better coefficient of friction. It is designed as a lining material with superior quality, so it is ideal for material flow problems.

UHMWPE 2000 – Dry Run 912

A superior UHMW Polymer. This special material was designed for the packaging and machine industry. Excellent for use in dry running conveyor applications with the antistatic filler reducing static build up on fast moving equipment.

UHMWPE 2000 – F-Ex #999

A virgin UHMW polymer with advanced flame retardant properties. This material is halogen free. Even with the additive used to extinguish the flame, great gliding properties and good wear resistance are still achieved through using the highest polymer possible. With its high impact strength, it was designed for such applications in conveyor systems used in the mining field. The flame retardant classification is measured to UL (Underwriters Laboratories) standard using the method UL94 = V0, vertical flame test.

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Also known as

- UHMWPE or Ultra High Molecular Weight Polyethylene

Similar to

- Okulen™ Poly Hi Solidur™
- Tivar™

UHMWPE 2000 – GB

Enhanced with a glass filler which gives the UHMW a higher abrasion resistance than standard material. It offers excellent impact strength for tough applications. The 983 is coloured black for outdoor use and it is UV-stabilized. It is also available in Green and Brown.

UHMWPE 2000 – H 925

The premier UHMW Polymer containing a unique antioxidant which protects the material from breaking apart at extreme temperatures. This material grade is used in high temp applications where normal UHMW fails. The enhanced material was designed to still operate effectively under lower load, with a temperature range peaking 150°C for a short time.

UHMWPE 2000 – MO

A material with high abrasion resistance and excellent sliding properties. Because of the low coefficient of friction this material allows even lower transmission power.

UHMWPE 2000 – L 845

An oil filled UHMW Polymer. This material was designed for the conveying industry. The enhanced lubricated oil filler helps on high speed chains or heavy dry run areas of the conveyor. Further, it helps reduce noise levels with its excellent gliding properties. This UHMW is known for its great impact strength and abrasion resistance surface quality. This material is FDA compliant for food applications.

UHMWPE 2000 GRM-956

A further developed product based on OK 2000 MR-963. This material has been developed for high tech paper-making machines. It is a sintered ultra-high molecular weight polyethylene (UHMW-PE, 9,2 mio. g/mol) Polymer with molybdenum disulphide and UV-stabilizers. A special graphite additive was included to get the friction values on as low a level as possible.

Properties

Property	Unit of measure	Value
Tensile Strength	Mpa	17
Elongation at Break	%	>50
Modulus of Elasticity	MPa	680
Impact Strength	kJ/m ²	200
Compressive Strength	MPa	
Shear Strength	MPa	
Shore Hardness	D Scale	64
Co-efficient of Friction	μ	0.12

Property	Unit of measure	Value
Water Absorption @ 23 Deg C	%	0.01
Service Temperature Intermittent	°C	120
Service Temperature Continuous	°C	80
Melting point	°C	135
Thermal Conductivity	W/m.°C	0.4
Dielectric Strength	kV/mm	
Volume Resistivity	Ω.cm	>10 ¹⁴
Surface Resistivity	Ω	>10 ¹³
Arc Resistance	Sec	